Question ID: cdec4c87

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Easy

Question

$$y = 12x - 20$$

$$y = 28$$

What is the solution (x, y) to the given system of equations?

Answer

A. $(4, 28)$

B. $(20, 28)$

C. $(28, 4)$

D. $(28, 20)$

Correct Answer: A

Rationale

Choice A is correct. The second equation in the given system is $y = 28$. Substituting 28 for y in the first equation in the given system yields $28 = 12x - 20$. Adding 20 to both sides of this equation yields $48 = 12x$. Dividing both sides of this equation by 12 yields $4 = x$. Therefore, the solution (x, y) to the given system of equations is $(4, 28)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the solution (y, x), not (x, y), to the given system of equations.

Choice D is incorrect and may result from conceptual or calculation errors.